Computational modelling of salmon migration in the Tornionjoki river

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1. Introduction

The amount of migrating salmon to the Gulf of Bothnia's rivers such as Tornionjoki, Kalixjoki and Simojoki has decreased in recent years which is concerning.[2] The reason for the sharp decrease in migrating salmon is not yet known.[3]

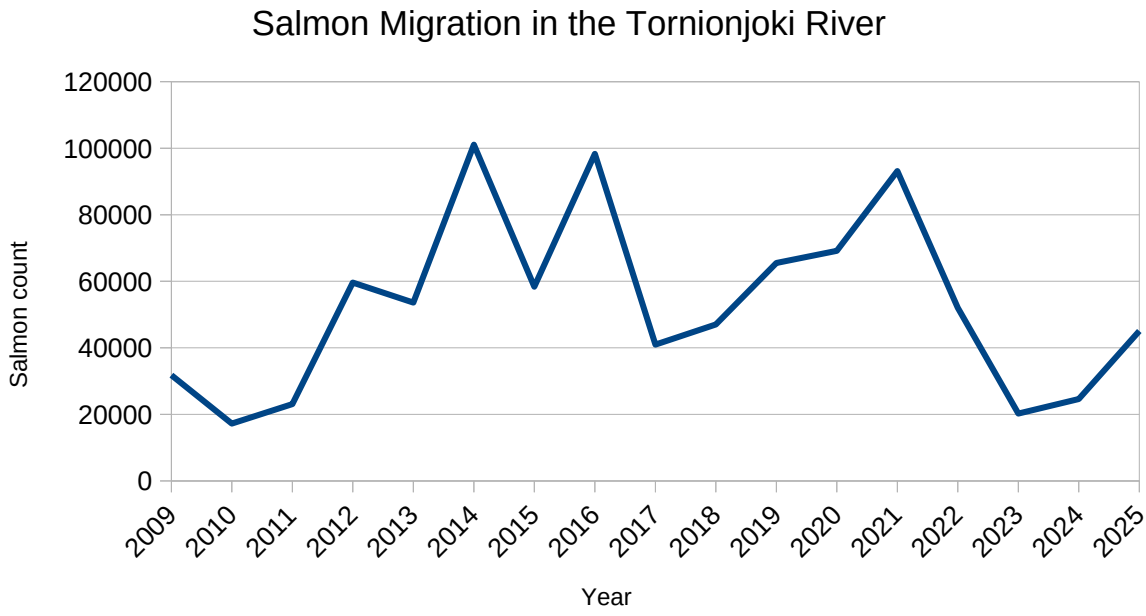


Figure 1: *Yearly salmon migration of the Tornionjoki river.[1]*

Tornionjoki is Western Europe's longest free-flowing river and its an important river for professional and recreational fishing.[4] Figure 1 shows how salmon migration has significantly decreased in the years 2023 and 2024. Salmon migration increased in the summer of 2025, but most of the salmon were male, which don't produce more young salmon.[5]

The fishing tourism of Tornionjoki has been estimated to be worth around 10.8 million euros.[6] Tornionjoki river has a fishing culture important to the locals.[7]

The study aims to examine why salmon migration has sharply declined in recent years by creating a computational model to represent the fluctuations of migrating salmon for the years 2009–2025. (Figure 1). The model will be used to predict future salmon migration and estimate the impact of fishing and seals on salmon migration.

2. Salmon life cycle

In the study the Atlantic salmon (*Salmo salar*) life cycle is modelled after Tuomas Manelius's Master's thesis.[4] Atlantic salmon are an anadromous species whose life cycle begins by migrating into the sea from the river and then returning back to their home river to spawn after feeding in the sea. Salmon alevin are born during spring. In the next stage the young salmon start to search for food. The mortality of young salmon called parr in this stage is high due to illnesses, predators and the shortage of food. After this stage salmon mortality isn't as high. The river stages of Atlantic salmon lasts 2–5 years in the Tornionjoki river.[4]

The young salmon start to undergo different physiological and behavioral changes after the river stages. This is called smoltification. In this stage the young salmon prepare for the sea. The migration of young salmon called smolts happen during spring and the early summer. Smolts from the Tornionjoki river swim to the main basin and southern parts of the Baltic sea. After one winter at sea some of the male salmon achieve maturity. On average Atlantic salmon feed in the sea around one to four years, but can feed for longer.[4] Fishing and seals cause salmon mortality in the sea.[8]

Large salmon ascend their home river during spring and the early summer. The first sea-winter salmon ascend during the period between July and August. Salmon spawn during September to November. After spawning, some of the salmon migrate back to the sea, while others stay in the river for the winter and return to the sea the following spring.[4] Most of the salmon can't endure the stress of spawning and die after.[9]

3. Material and methods

3.1 Statistics on salmon runs, fisheries, and seal populations

Salmon migration data and gray seal (*Halichoerus grypus*) population data were collected from the database of the Natural Resources Institute Finland (Luke). [1], [10] Marine fishing statistics were collected from an ICES (International Council for the Exploration of the Sea) report, table 2.2.1.4, from the coastal and offshore fisheries columns for the Baltic Sea main basin and the Gulf of Bothnia.[11], [12] The river fishery data of the Tornionjoki river were collected from a Swedish-Finnish report.[13] Missing values from the datasets were computationally estimated after which the collected statistics were tabulated and used as the model's input. (Appendix 1)

3.2 Simplified model

The simplified state-based model of the Tornionjoki river's salmon life cycle is split into three different phases: freshwater, sea-phase salmon and spawning salmon. The freshwater phase lasts 2–5 years in the Tornionjoki river and each sea-phase and spawning phase lasts for one year.

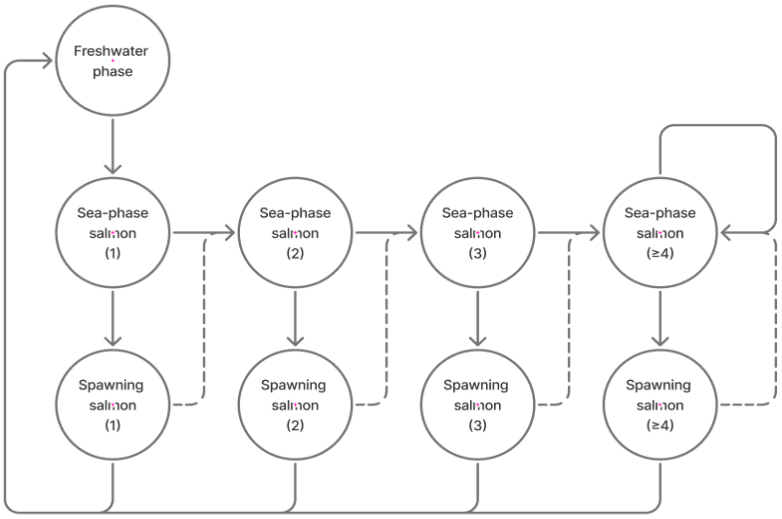


Figure 2: *Simplified state-based model of the Tornionjoki river's salmon life cycle.*

The mortality factors were assumed to be yearly gray seal populations and fishing. The mortality rates of freshwater salmon and spawning salmon were represented as transition probabilities in the state-based model. The transition probabilities of the state-based model were collected from previous studies.[9] Transition probabilities that weren't found from previous studies were estimated computationally.

3.3 Computational model

A computational model was written in the Rust programming language to describe salmon population fluctuations and survival. The model's code is Appendix 10. The ndarray crate for Rust was used for matrices and their mathematical operations.[14], [15]

A Leslie population model was used for modelling the salmon runs. Let $\mathbf{n}_t$ be the salmon population vector at time t and **L** be the Leslie matrix. Thus the Leslie model is $\mathbf{n}_{t+1} = L\left(\mathbf{n}_t - \alpha\mathbf{H}_t - \mathbf{K}_t\right)$.[16], [17]

$$\begin{bmatrix} n_0 \\ n_1 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_{t+1} = \begin{bmatrix} f_0 & f_1 & \dots & f_{\omega-2} & f_{\omega-1} \\ s_0 & 0 & \dots & 0 & 0 \\ 0 & s_1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & s_{\omega-2} & 0 \end{bmatrix} \left(\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_{\omega-1} \end{bmatrix}_t - \alpha \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_{\omega-1} \end{bmatrix}_t - \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_{\omega-1} \end{bmatrix}_t \right)$$

Formula 1: *Matrix representation of the Leslie model*

The first row of the Leslie matrix has the fecundities of the population. The other elements of the matrix contain the transition probabilities from one age class to another.

One year was used as the Leslie model's time step. The program has two population vectors, one for males and one for females. It is assumed that the sex ratio of eggs is even and the Leslie matrix of females have fertility rates, which the computational model splits into males and females in a 1:1 ratio.

Before the state transition the population vector $\mathbf{n}_t$ is decreased by the fishing amount, which includes river fishing and sea fishing, and the mortality produced by gray seals. The mortality caused by gray seals is assumed to be directly proportional to the seal population size. This mortality equals the population multiplied by the constant parameter alpha. Mortality caused by seals and fishing is distributed proportionally to the number of males and females. Because there are more gray seals in the Bothnian Bay compared to the Baltic Sea's main basin, seal induced mortality is split between sea-phase salmon and spawning salmon.[8] Eighty percent of the mortality caused by seals is split to spawning salmon and twenty percent to sea-phase salmon because there are more seals near the Tornionjoki river's mouth and they eat more salmon.[8]

Before sea fishing is applied to the Leslie model the program multiplies it by the factor below:

$$\text{Sea fishing weight} = \frac{L_s}{L_s + \gamma} \quad ,$$

Formula 2: *Sea fishing is multiplied by the proportion of Tornionjoki salmon in the sea.*

where L_s is the amount of Tornionjoki salmon in the sea and γ is the amount of salmon from other stocks in the Baltic sea. As sea fishing is directed at all salmon stocks in the Baltic sea only a portion of the salmon are from the Tornionjoki river's stock. Therefore it is presumed that the amount of salmon from other stocks is constant to simplify the model.

The age distribution of the model's salmon population in the beginning is estimated by calculating the stable age structure of the Leslie matrix. This is done by simulating the Leslie model for 100 years without fishing and seal

induced deaths. The resulting population vector is normalized and scaled to correspond with first year salmon run data. Then the program runs the Leslie model once with year one data to simulate seal induced unbalance in the age distribution.

The first year data of fishing was left out of the input because it differed from later years. Analysis showed that the first year fishing data caused instability to the model. Therefore due to the analysis and the lack of historical data, the input was narrowed to later, more comparable years.

3.4 Fitting the computational model

The parameters of the population model were fitted to the observation data of the Tornionjoki river's salmon runs from years 2009 to 2025. The parameters were chosen by maximizing the coefficient of determination between the calculated salmon runs and the measured salmon run data.[18] The predictions of the parameter combinations that gave the best coefficient of determinations were checked manually. The parameters that matched realistic values were chosen for the final computational model. Some parameter combinations that were able to give a high coefficient of determination had predicted that salmon migration would be zero in the year 2030. These parameter combinations resulted in a much lower abundance of salmon from other stocks in the Baltic Sea (Gamma value in Formula 2) compared to other combinations, which produced more realistic values.

4. Results

The result was a population model that describes the fluctuations of the Tornionjoki river's salmon migration in the years 2009–2025. Future years were predicted using the model and the impact of seals and fishing were assessed.

4.1 Final computational model

Table 1: *Estimated parameters of the computational model.*

Parameter	Description	Estimated value
$\mathbf{P}_L$	The probability that freshwater phase salmon migrate to the ocean each year. Young salmon migrate to the ocean after 2–5 years.	0.1
$\mathbf{P}_J$	The probability that a salmon stays in the ocean and doesn't ascend the river every year.	0.2
Alpha (α)	Scales the mortality produced by seals in the Leslie model.	0.6
Gamma (γ)	The amount of salmon from other stocks in the Baltic sea. Scales sea fishing.	420000
Fecundity	How many eggs a female salmon lays.	600

Transition probabilities $\mathbf{P}_L$ and $\mathbf{P}_J$ were chosen as parameters, because similar values were not found from previous studies. The mortality caused by gray seals was estimated using the alpha parameter. The gamma parameter was chosen to calculate sea fishing because the Baltic sea has salmon from other river stocks and hatchery-reared salmon. Fecundity was chosen as a constant parameter to simplify the model.

The model can follow the Tornionjoki river's salmon population fluctuations but it has significant error in the beginning. This may be due to a lack of historical data, which reduces the model's predictive capability during the initial years of the measured data (Figure 3).

4.2 The model's prediction of salmon migration until the year 2030

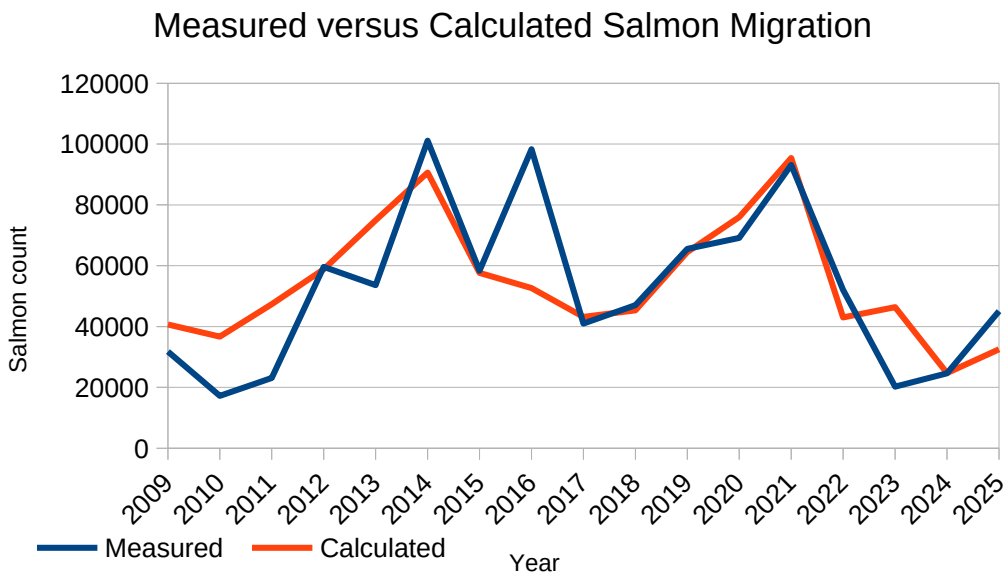


Figure 3: *The calculated salmon runs compared to the measured salmon runs in the Tornionjoki river for the years 2009–2025.*

The computational model was used to predict salmon migration until the year 2030. Variation of either the number of gray seals or fishing was done in Figures 5 and 6 while the other value was constant. The exact values of the variation are in Appendix 2. The lightest curve represents the most amount of seals or fishing and the darkest curve represents the least amount of seals or fishing in Figures 5 and 6.

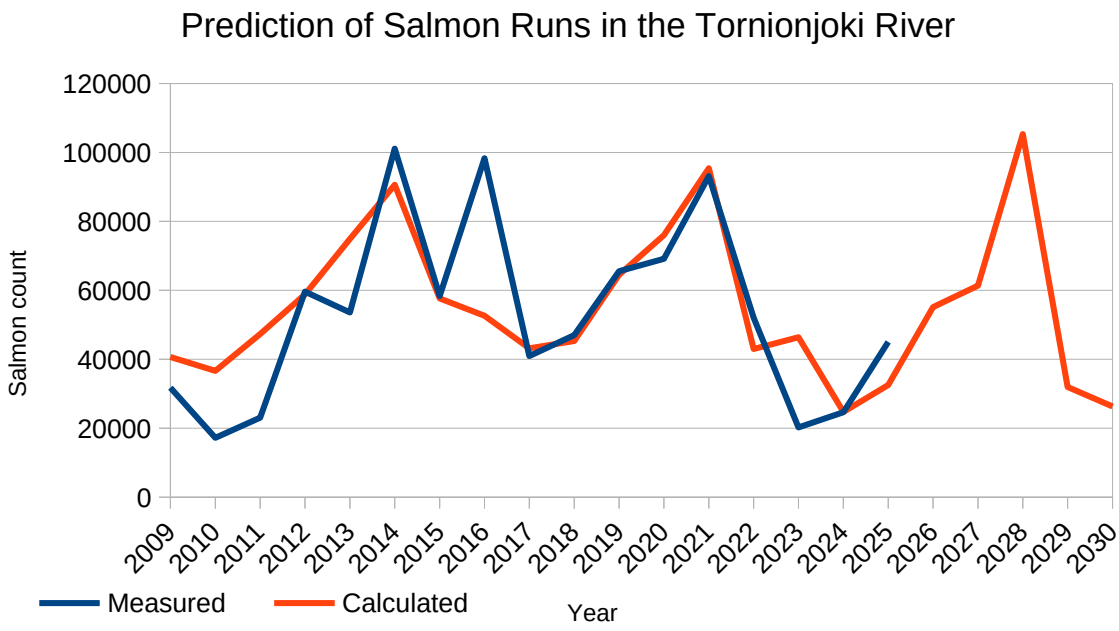


Figure 4: *The main prediction of the Tornionjoki river's future salmon runs.*

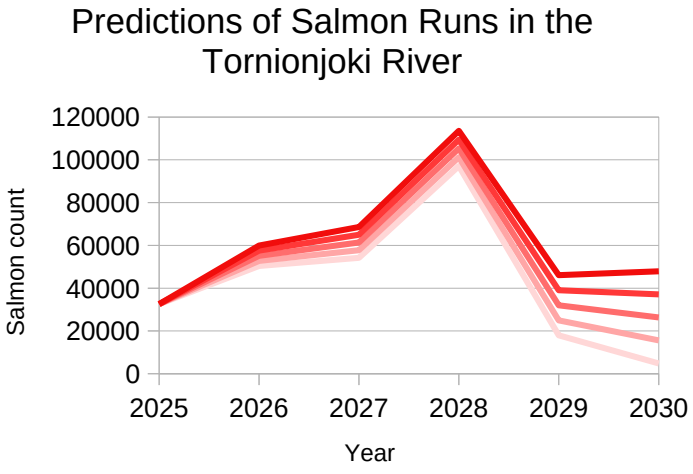


Figure 5: *Variation in the number of gray seals.*

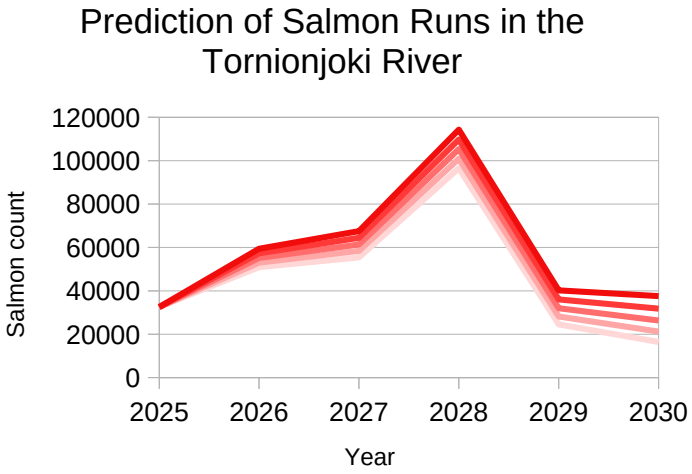


Figure 6*: Variation in fishing.*

5. Discussion

The model observes a delay between good salmon years and predicts good salmon years for 2027 and 2028 followed by a crash at the end of the decade caused by the recent weak salmon years' low production of young salmon. The model predicts a similar curve compared to the measured data for the years 2023–2025 with a one year delay in Figure 4. This might be because of 2025's exceptional amount of young male salmon.[5] According to the prediction while taking into

account the one year delay, the Tornionjoki river's salmon migration will increase in 2026 and peak at 2027.

People can affect salmon migration by reducing fishing and hunting gray seals. The effects of these have been estimated by the predictions in Figures 5 and 6. According to the model the effect of fishing and gray seal population changes can be seen with a delay of five years.

5.1 Error analysis of model

5.1.1 Sensitivity analysis

A sensitivity analysis was made for the computational model by calculating results while the inputs were varied. The variation was done using a nearly orthogonal Latin hypercube (Appendix 4).[19] The calculated results for each parameter were gathered into scatter plots. (Appendices 5–9)

The result was that the model is very sensitive to its parameters. Change in fecundity either leads to exponential growth or decline in salmon migration. According to the computational model the removal of both fishing and seals leads to strong growth to biologically impossible levels. However the salmon population's growth is limited by other biological factors such as the quantity of food when seals and fishing are removed.

5.1.2 Assessment of model reliability

In addition the results of the sensitivity analysis the computational model lacks historical data and its accuracy is based on only 15 years of measured salmon migration data. The model also has many presumptions such as a constant fecundity that decrease the accuracy of the computational model.

5.2 Future development of the model

Biological constraints such as food availability should be added to the model. More historical data should be accumulated to be able to create a better model and gender related differences should be considered. The model could also be integrated into a larger framework to study relationships such as those between gray seals and Atlantic salmon.

6. Conclusions

The computational model was able to explain the population fluctuations of 2017–2022 well, but it had significant error with earlier years. According to the model the weak salmon migration in recent years is because of natural population

fluctuations. The low production of young salmon of the previous weak years can be seen with a delay in 2023 and 2024. According to the prediction in Figure 4 and taking into account the one year shift, Tornionjoki salmon migration will peak during the year 2027 and then decline. According to Figures 5 and 6 gray seals and fishing affect salmon migration with a delay of five years.

7. Acknowledgements and use of AI

The AI-assisted LaTex editor Prism by OpenAI was used for creating the latex code for the formulas of this report.[20] The OpenAI's GPT-5.5 large artificial intelligence language model was used for converting Rust vectors to LibreOffice Calc tables and vice versa.[21] The results were verified by hand.

Different ways of creating the state transition matrix were discussed and the grammar of the original Finnish version was checked with my teachers Esa Lappi and Merikki Lappi.

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